Этап 1 — подсчёт частот предметов

BREADBASKET — bakery transactions

Транзакций: 9465 | MinSupport: 0.5% | Минимальная частота: 48

Все часто встречающиеся предметы остаются в FP-дереве.

Предмет	Частота	Поддержка
Coffee	4528	47.84%
Bread	3097	32.72%
Tea	1350	14.26%
Cake	983	10.39%
Pastry	815	8.61%
Sandwich	680	7.18%
Medialuna	585	6.18%
Hot chocolate	552	5.83%
Cookies	515	5.44%
Brownie	379	4.00%
Farm House	371	3.92%
Juice	365	3.86%
Muffin	364	3.85%
Alfajores	344	3.63%
Scone	327	3.45%
Soup	326	3.44%
Toast	318	3.36%
Scandinavian	275	2.91%
Truffles	192	2.03%
Coke	184	1.94%
Spanish Brunch	172	1.82%
Baguette	152	1.61%
Tiffin	146	1.54%
Fudge	142	1.50%
Jam	142	1.50%
Mineral water	134	1.42%
Jammie Dodgers	125	1.32%
Chicken Stew	123	1.30%
Hearty & Seasonal	100	1.06%
Salad	99	1.05%
Frittata	81	0.86%
Smoothies	77	0.81%

Этап 2 — сортировка транзакций

Предметы упорядочены по убыванию общей частоты, до построения дерева

T01 Bread

T02 Scandinavian

T03 Hot chocolate → Cookies → Jam

T04 Muffin

T05 Coffee → Bread → Pastry

T06 Pastry → Medialuna → Muffin

T07 Coffee → Tea → Pastry → Medialuna

T08 Bread → Pastry

T09 Bread → Muffin

T10 Medialuna → Scandinavian

T11 Bread → Medialuna

T12 Coffee → Tea → Pastry → Jam

T13 Coffee → Bread

T14 Bread → Pastry → Medialuna

T15 Scandinavian → Mineral water

T16 Coffee → Bread → Medialuna

T17 Hot chocolate

T18 Farm House

T19 Bread → Farm House

T20 Bread → Medialuna

T21 Coffee → Bread → Medialuna

T22 Jam

T23 Muffin → Scandinavian

T24 Bread

T25 Scandinavian

T26 Fudge

T27 Scandinavian

T28 Coffee → Bread

T29 Bread → Jam

T30 Bread

В следующих страницах показано FP-дерево после добавления каждой транзакции (или контрольных точек).

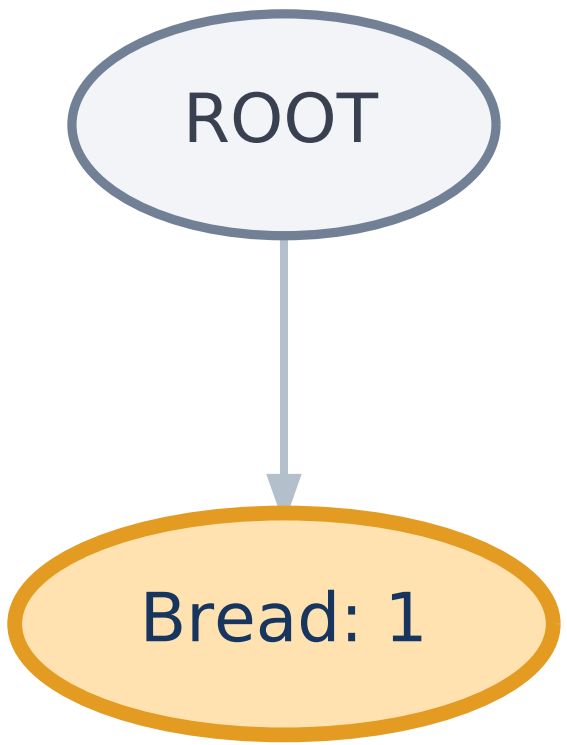
Этап: пустое дерево

Before inserting any transactions



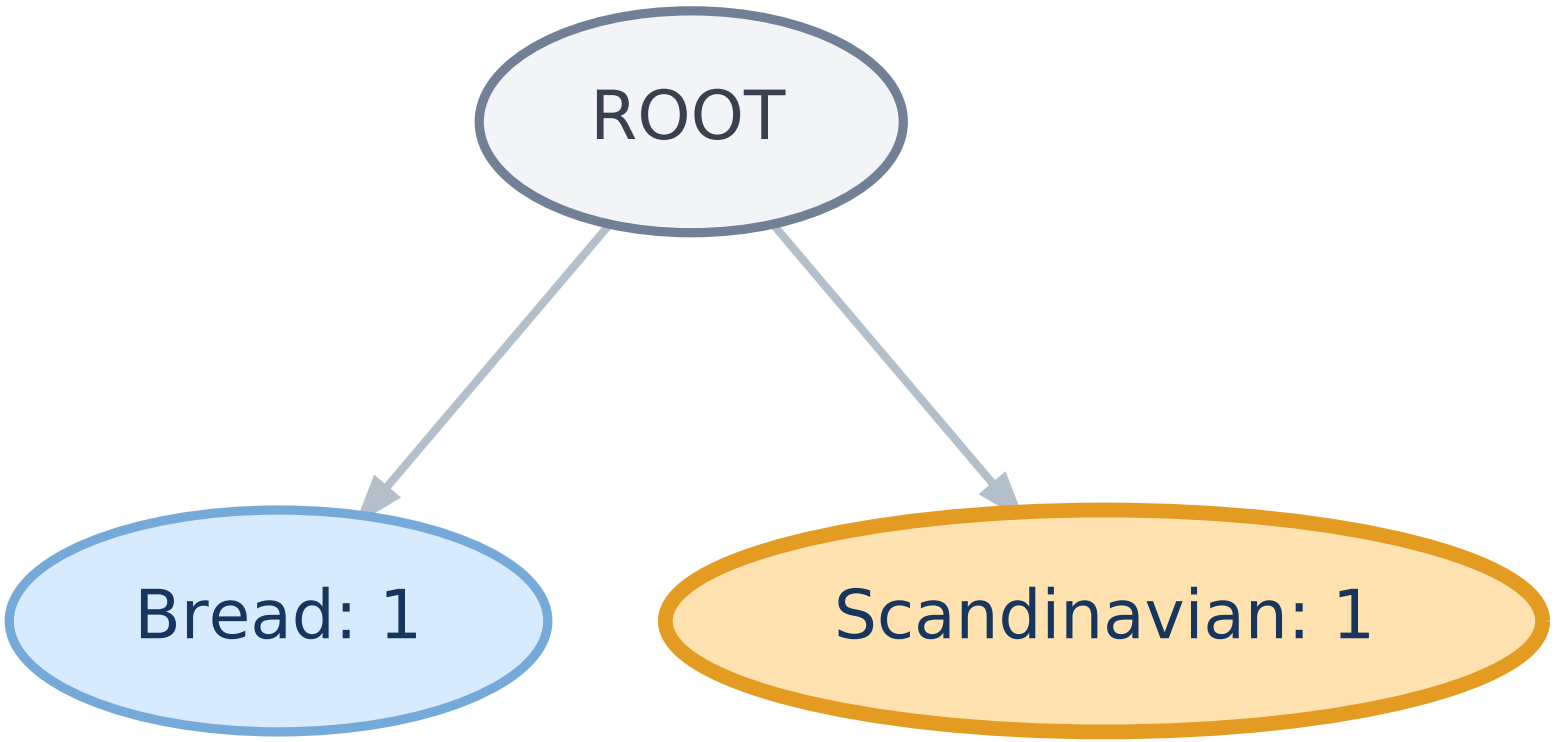
Этап: после транзакции 1 из 9465

Added path: Bread



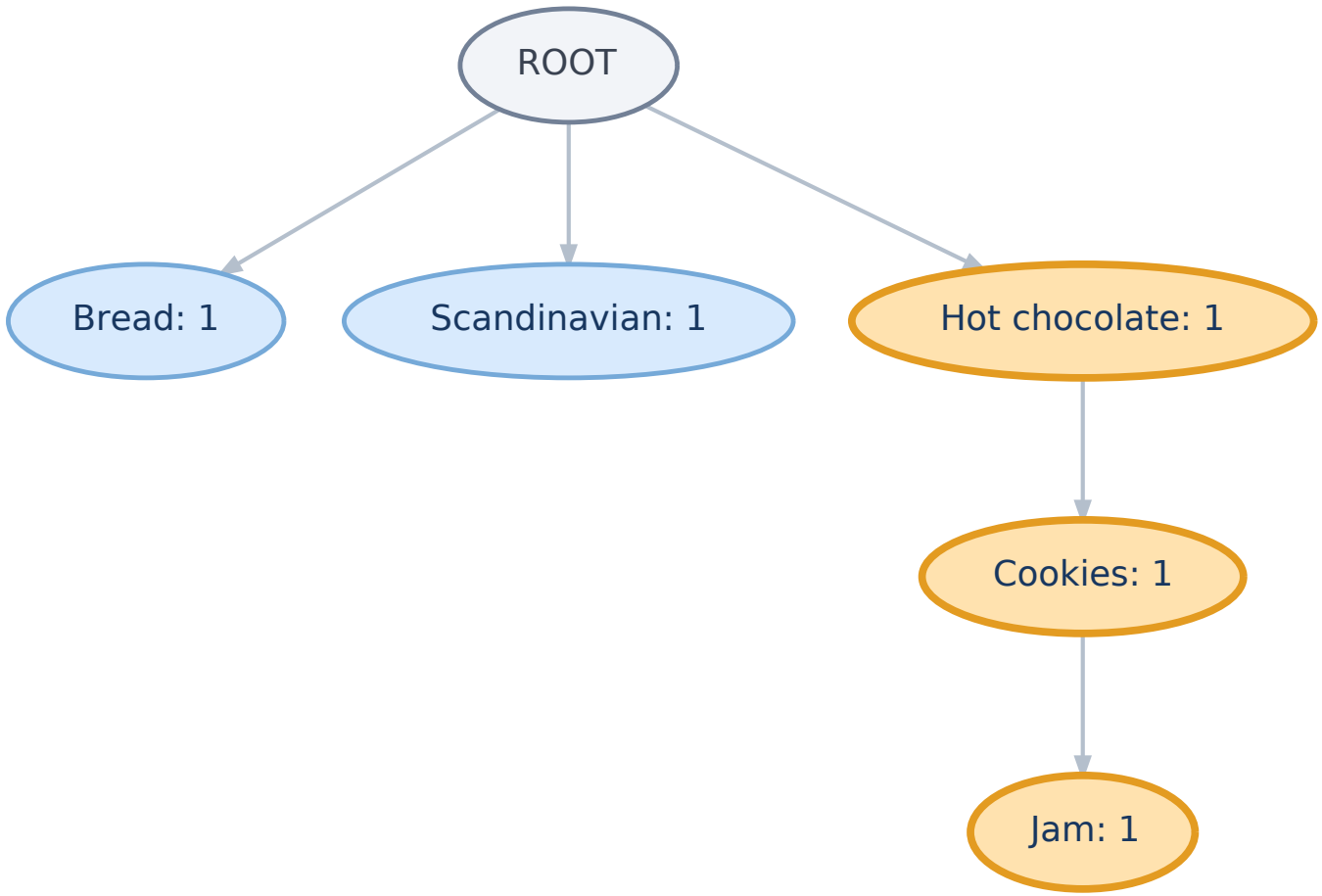
Этап: после транзакции 2 из 9465

Added path: Scandinavian



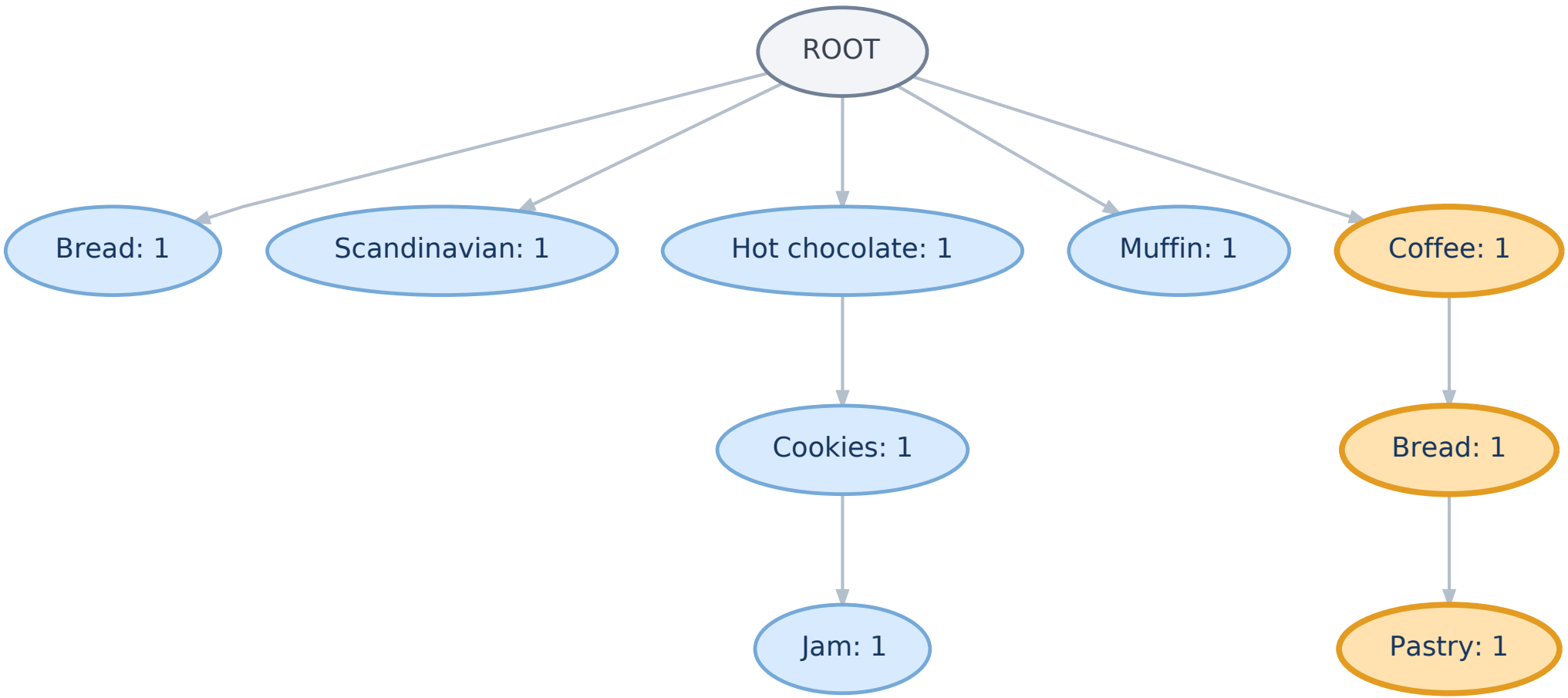
Этап: после транзакции 3 из 9465

Added path: Hot chocolate → Cookies → Jam



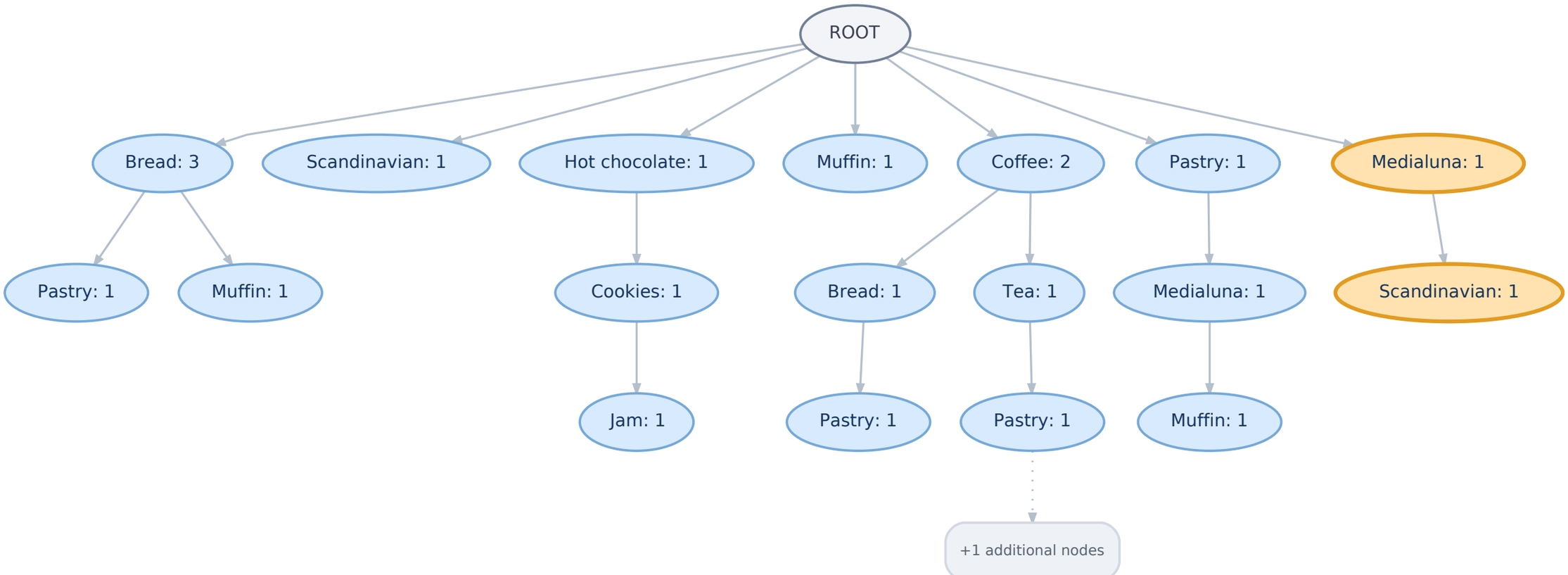
Этап: после транзакции 5 из 9465

Added path: Coffee → Bread → Pastry



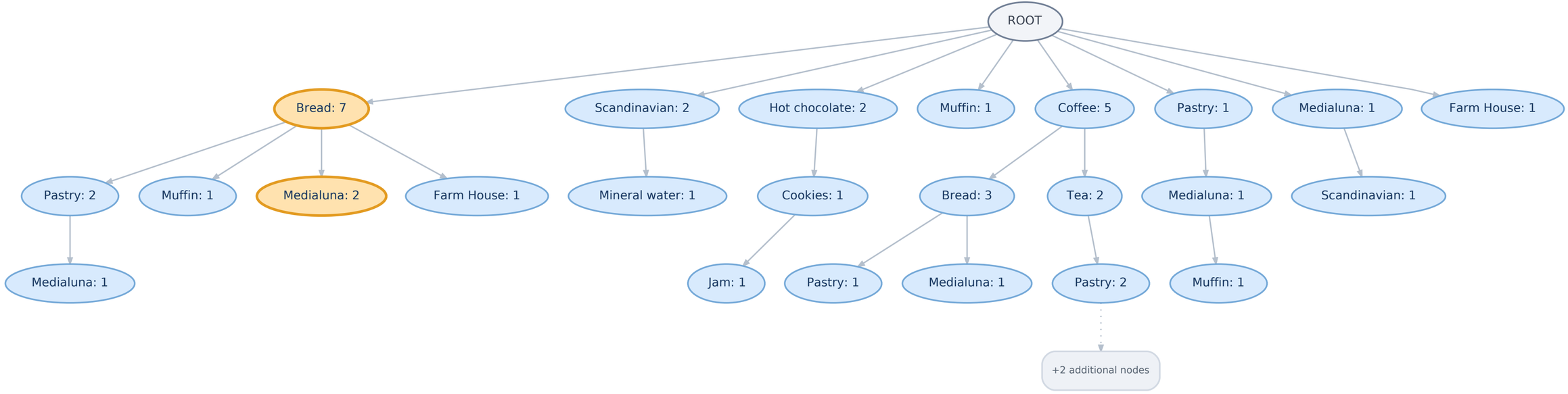
Этап: после транзакции 10 из 9465

Added path: Medialuna → Scandinavian



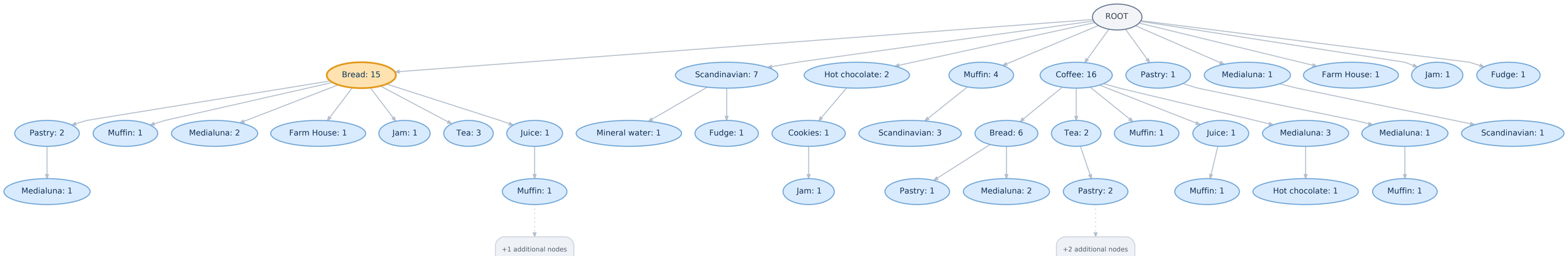
Этап: после транзакции 20 из 9465

Added path: Bread → Medialuna



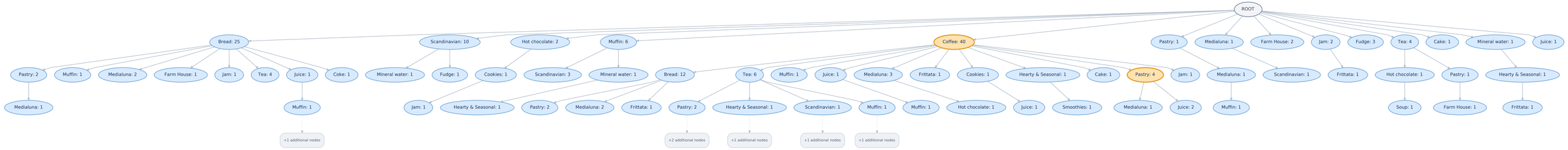
Этап: после транзакции 50 из 9465

Added path: Bread



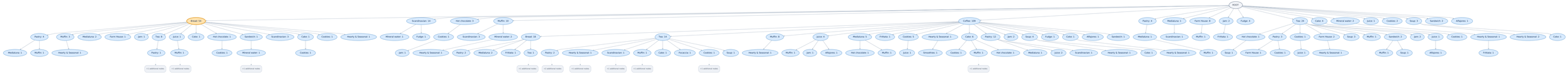
Этап: после транзакции 100 из 9465

Added path: Coffee → Pastry



Этап: после транзакции 250 из 9465

Added path: Bread



Этап: после транзакции 500 из 9465

Added path: Coffee → Sandwich



Этап: после транзакции 1000 из 9465

Added path: Bread → Keeping It Local



Этап: после транзакции 2000 из 9465

Added path: Tea



Этап: после транзакции 5000 из 9465

Added path: Coffee → Bread → Cake



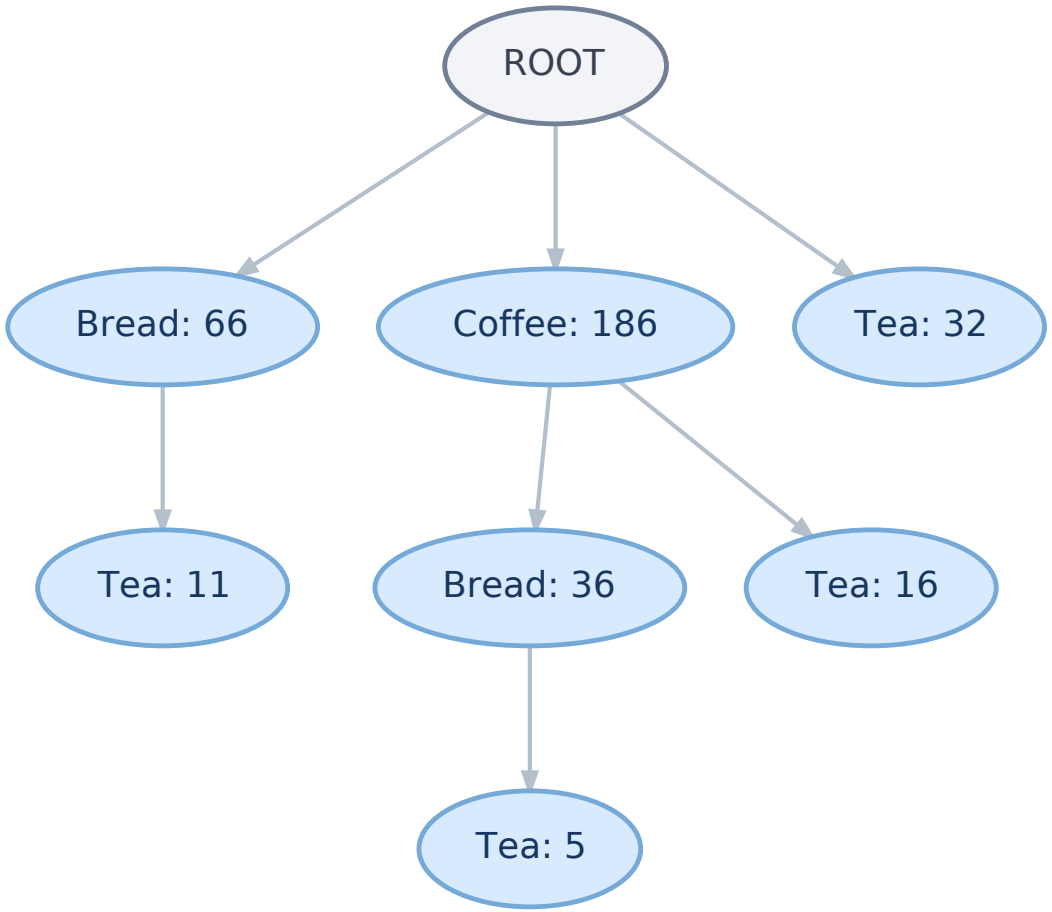
Этап: после транзакции 9465 из 9465

Added path: Smoothies



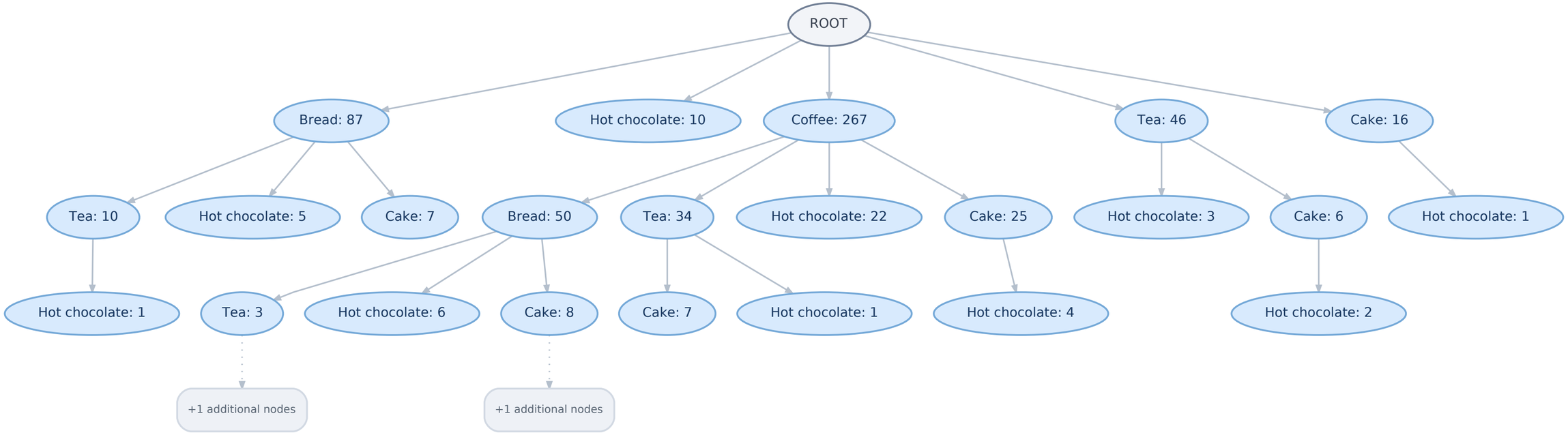
Условное FP-дерево: Brownie

Frequent prefix counts: Coffee=186, Bread=102, Tea=64



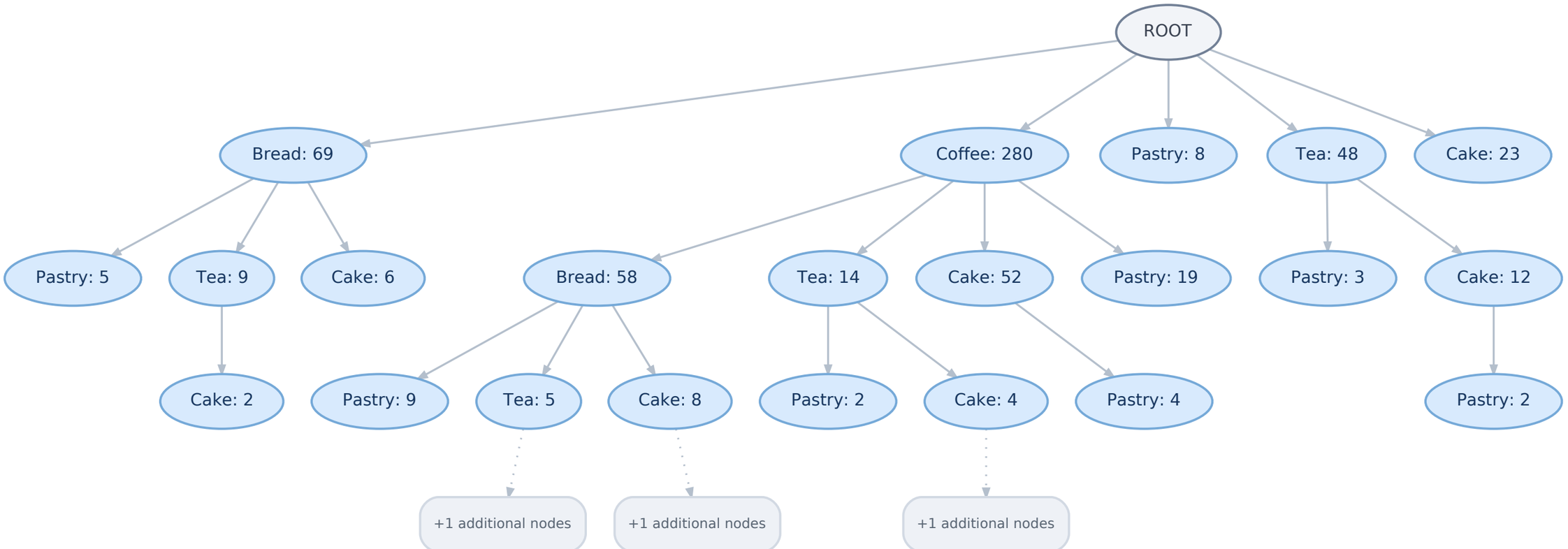
Условное FP-дерево: Cookies

Frequent prefix counts: Coffee=267, Bread=137, Tea=93, Cake=69, Hot chocolate=57



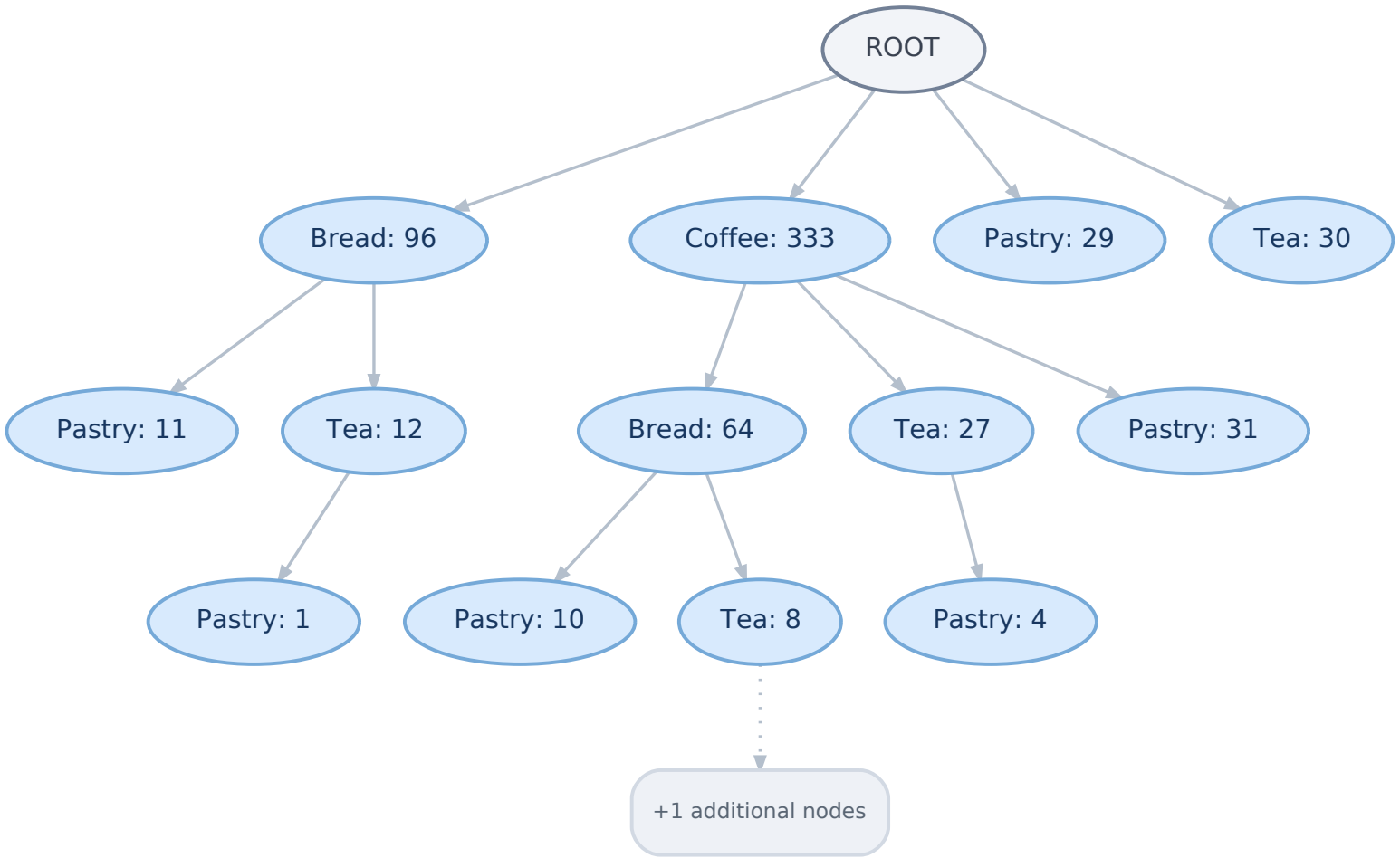
Условное FP-дерево: Hot chocolate

Frequent prefix counts: Coffee=280, Bread=127, Tea=76, Cake=108, Pastry=54



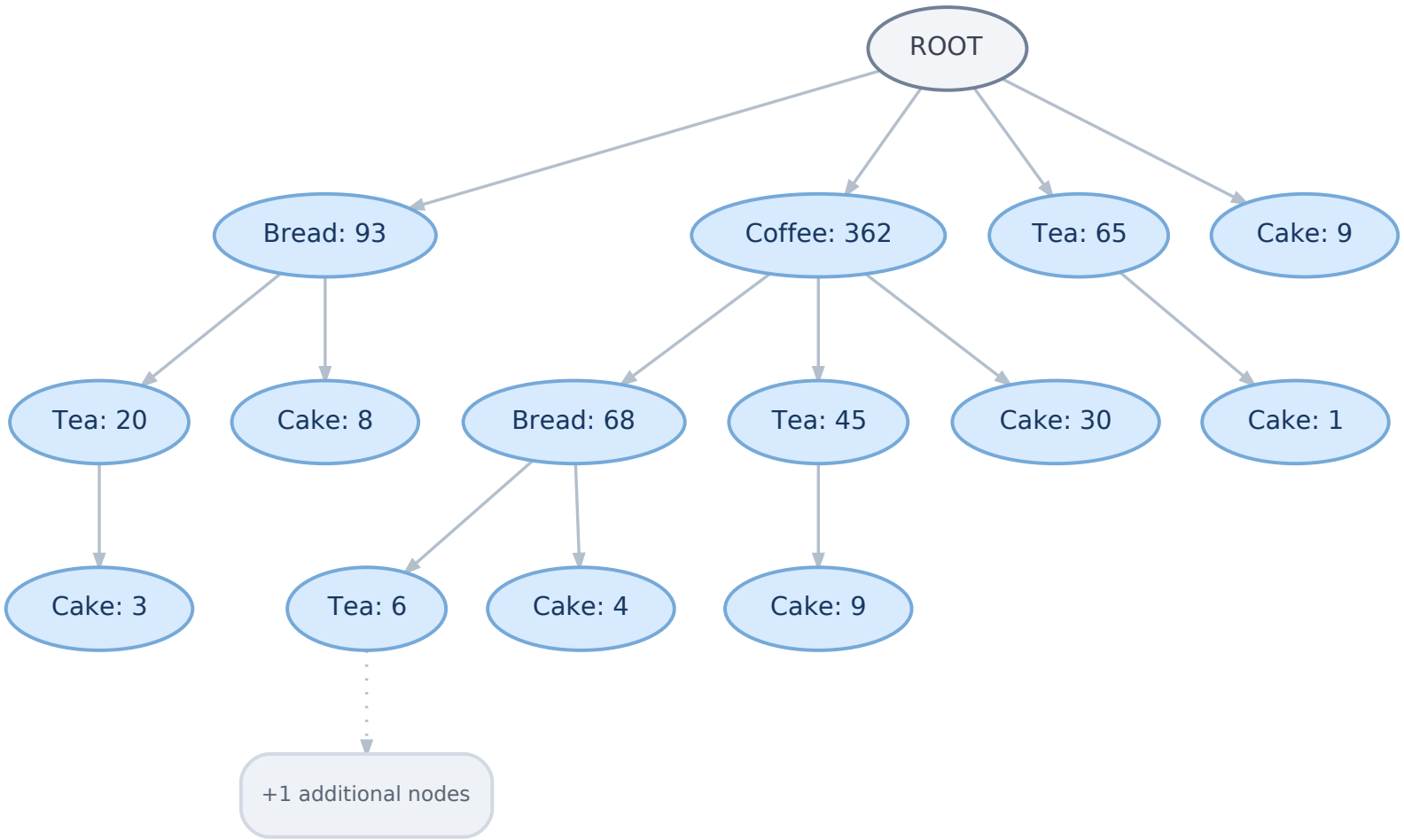
Условное FP-дерево: Medialuna

Frequent prefix counts: Coffee=333, Bread=160, Tea=77, Pastry=87



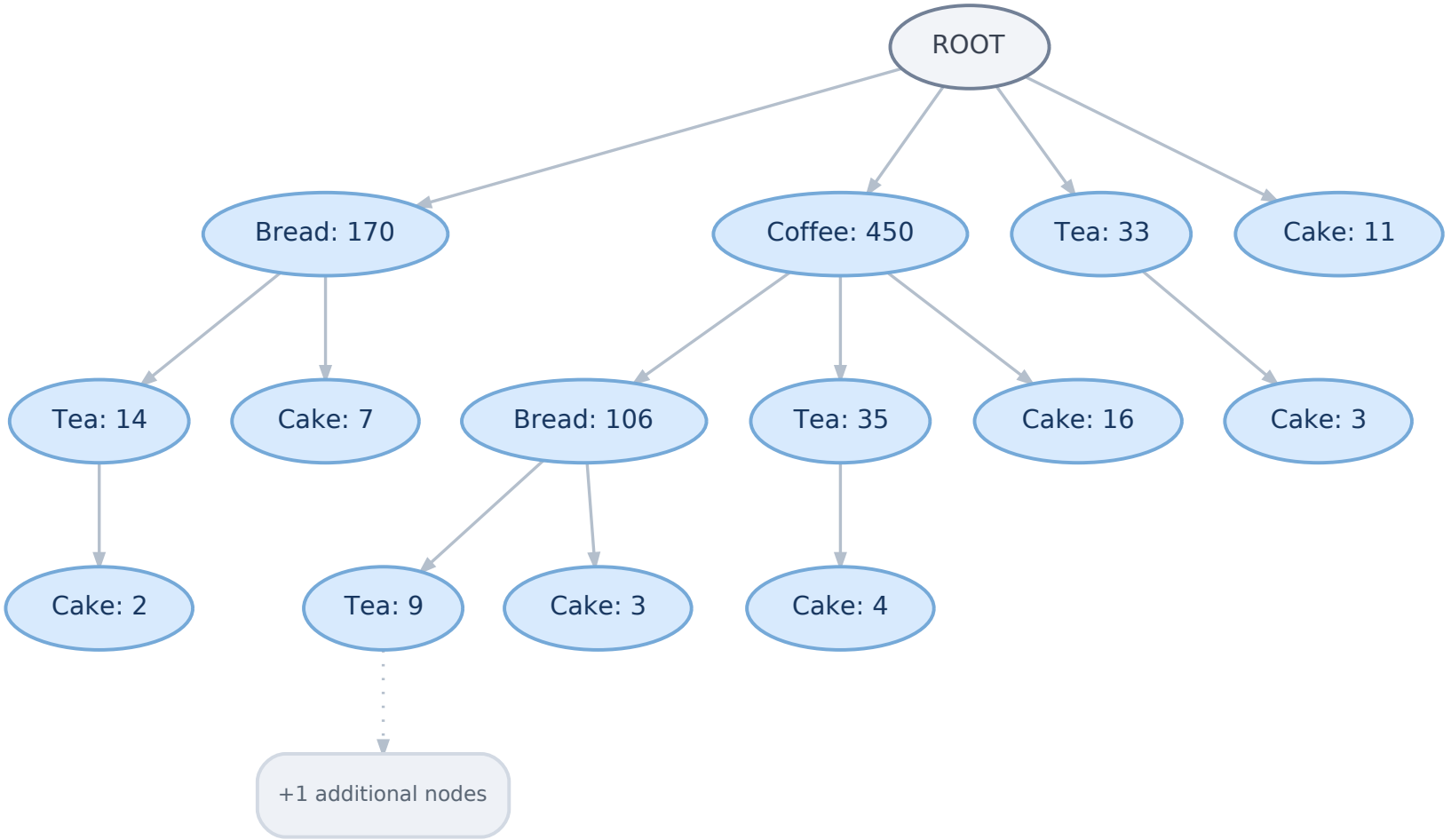
Условное FP-дерево: Sandwich

Frequent prefix counts: Coffee=362, Bread=161, Tea=136, Cake=65



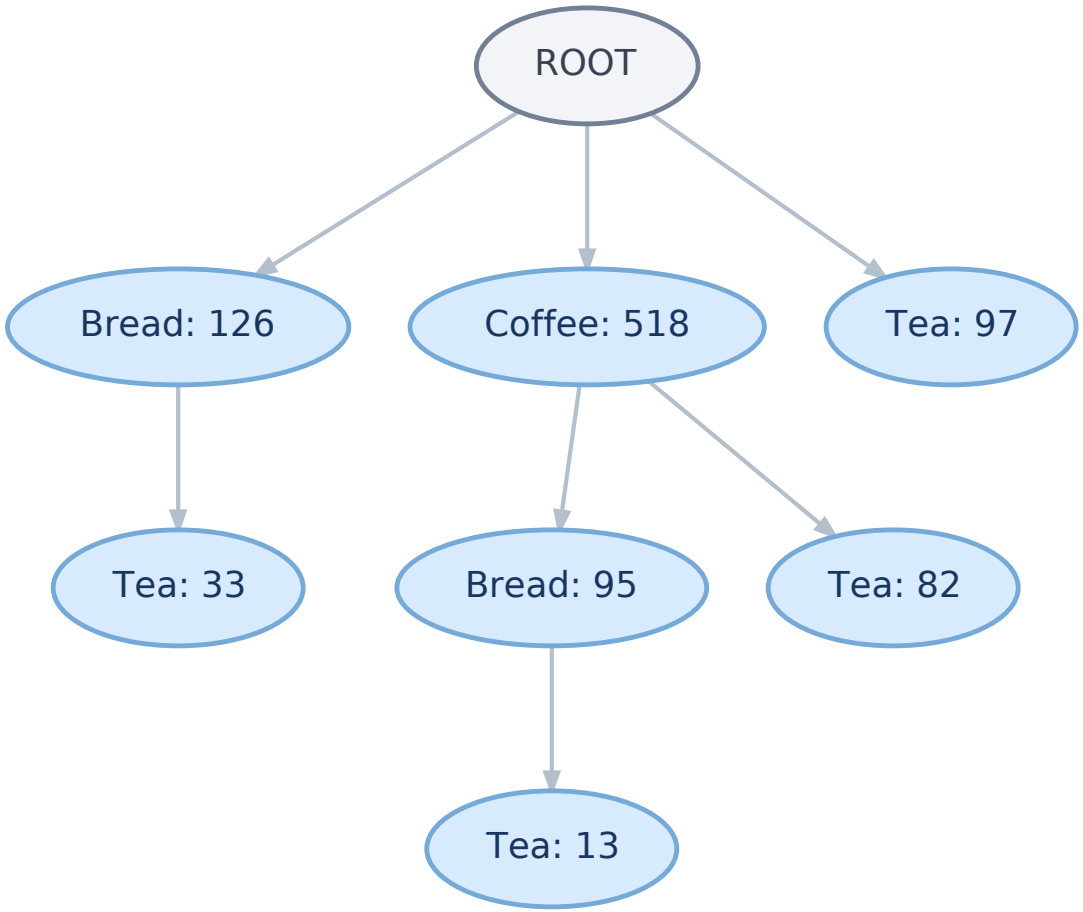
Условное FP-дерево: Pastry

Frequent prefix counts: Coffee=450, Bread=276, Tea=91, Cake=49



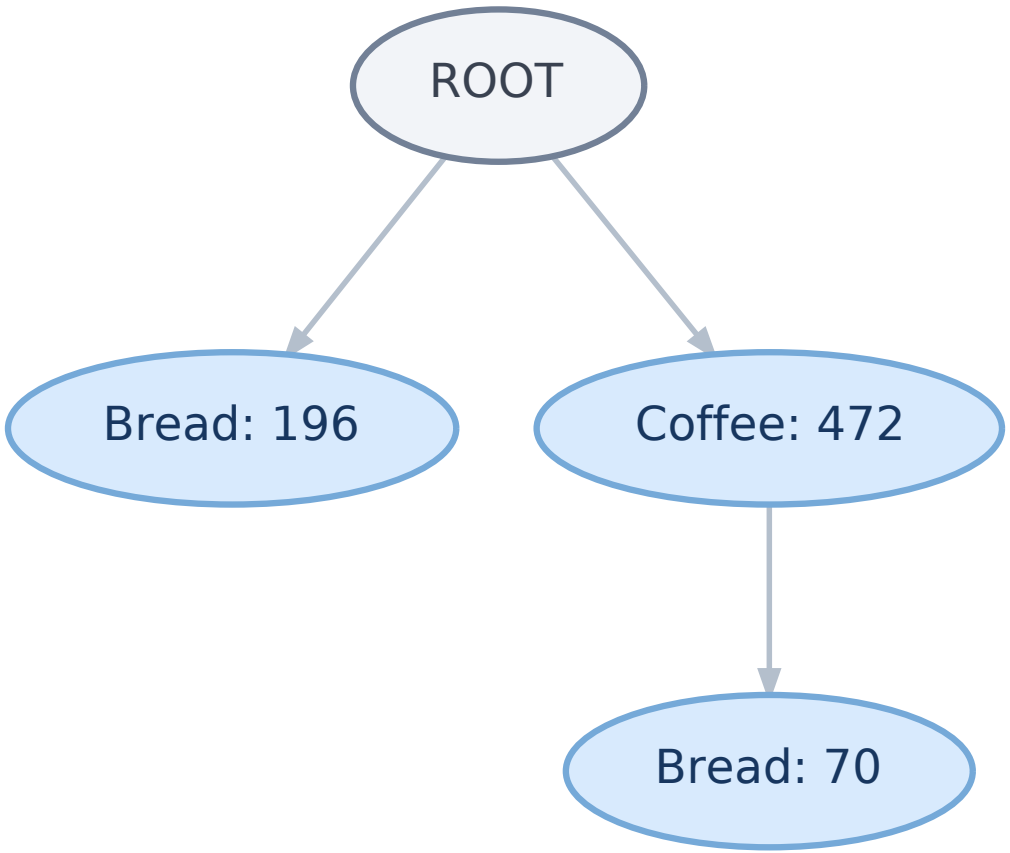
Условное FP-дерево: Cake

Frequent prefix counts: Coffee=518, Bread=221, Tea=225



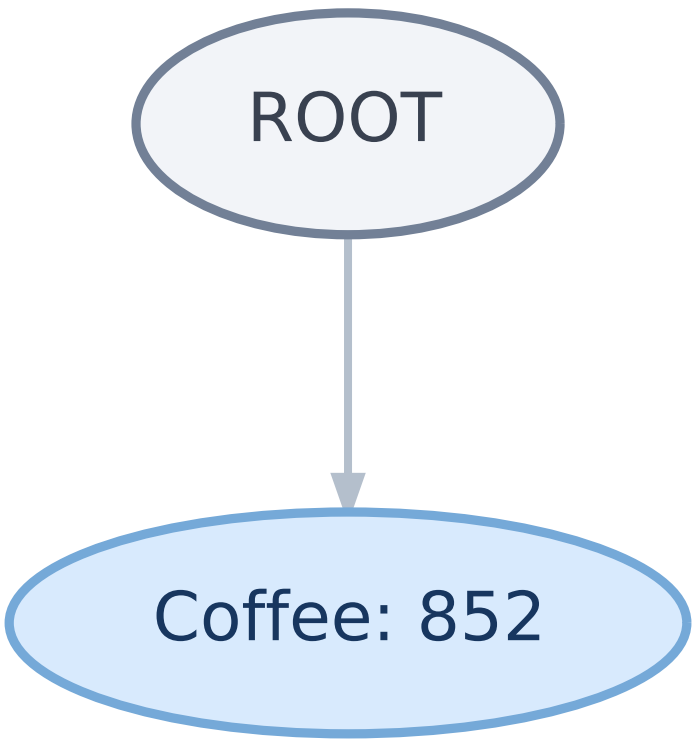
Условное FP-дерево: Tea

Frequent prefix counts: Coffee=472, Bread=266



Условное FP-дерево: Bread

Frequent prefix counts: Coffee=852



Условное FP-дерево: Coffee

Frequent prefix counts: No frequent prefix items



Итоги построения FP-дерева

BREADBASKET — bakery transactions

Узлов без ROOT: 2021 | Глубина: 9 | Частота: 48

Условные FP-деревья (для каждого суффикса)

Brownie: Coffee, Bread, Tea

Cookies: Coffee, Bread, Tea, Cake, Hot chocolate

Hot chocolate: Coffee, Bread, Tea, Cake, Pastry

Medialuna: Coffee, Bread, Tea, Pastry

Sandwich: Coffee, Bread, Tea, Cake

Pastry: Coffee, Bread, Tea, Cake

Cake: Coffee, Bread, Tea

Tea: Coffee, Bread

Bread: Coffee

Coffee: нет частых предшественников

Для полного набора правил используйте программную реализацию FP-Growth.